

Mark schemes

Q1.

(a) $\frac{dv}{dt} = 6t - 5$

M1 A1

2

(b) $6t - 5 = 0$

$$t = \frac{5}{6}$$

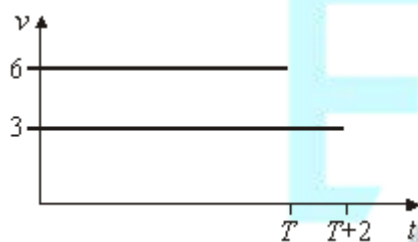
A1F

1

[3]

Q2.

(a)



horizontal line below other

(0, 3 to (T + 2, 3)

B1

B1

2

(b) (i) $x + 20$

B1

1

(ii) $x + 20 = 6T$

B1

1

(iii) $3(T + 2) = x$

M1A1

2

(iv) $6T - 3T - 6 = 20$

M1

$$3T = 26 \quad T = \frac{26}{3}$$

ft if T positive

A1F

$$\frac{26}{3} \times 6 = x + 20$$

M1

$$x = 32$$

ft if x positive

A1F

4

[10]

Q3.

(a) (i) $9 = 18 + 6a$

Use of constant acceleration equation to form an equation for a.

M1

$$a = -\frac{9}{6} = -1.5 \text{ ms}^{-2}$$

Correct equation

A1

ag *Correct acceleration from correct working.*

A1

3

(ii) $s = 18 \times 6 + \frac{1}{2} \times (-1.5) \times 6^2$

Use of constant acceleration equation to find s.

M1

$$= 81 \text{ m}$$

Correct distance

A1

2

(b) (i) $15 = 9 + 1.2t$

Use of constant acceleration equation to form an equation for t.

M1

$$t = \frac{15 - 9}{1.2} = 5\text{s}$$

Correct time

A1

2

(ii) $s = \frac{1}{2} (9 + 15) \times 5$

Use of constant acceleration equation to find s.

M1

Correct expression.

A1

$$= 60 \text{ m}$$

Correct distance.

A1

3

(c) $81 + 60 = 141 \text{ m}$

Correct sum of distances from (a)(ii) and (b)(ii)

B1

1

EXAM PAPERS PRACTICE

[11]

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Q4.

(a) $\mathbf{v} = -4e^{-t} \mathbf{i} + (6 - 3e^{-t}) \mathbf{j}$

Differentiating position vector

M1

$$t = 0 \text{ A1}$$

Correct velocity

$$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$$

ag Substituting t = 0 to obtain initial velocity

A1

3

(b) $\mathbf{a} = 4e^{-t} \mathbf{i} + 3e^{-t} \mathbf{j}$

Differentiating velocity

M1

Correct acceleration

A1

2

(c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Finding acceleration when $t = 0$

M1

$$a = \sqrt{4^2 + 3^2} = 5$$

Correct magnitude

A1

2

(d) $\mathbf{v} \rightarrow 0\mathbf{i} + 6\mathbf{j}$

For i component

B1

For j component

B1

2

[9]

Q5.

(a) $4^2 = 20^2 + 2 \times 48a$

Use of constant acceleration equation to find a

M1

$$a = \frac{16 - 400}{96} = -4 \text{ ms}^{-2}$$

ag Correct acceleration from correct working

A1

2

(b) $0^2 = 4^2 + 2 \times (-4)s$

use of constant acceleration equation to form equation for s

M1

$$s = \frac{16}{8} = 2 \text{ m}$$

Correct equation

A1

Correct s

A1

3

(c) $0 = 20 - 4t$

Use of constant acceleration equation to form equation for t

M1

$t = 5$

Correct t

A1

2

(d) $-F - 200 = 1100 \times (-4)$

Three term equation of motion

M1

Correct equation

A1

$F = 4200$

Correct F

A1

3

EXAM PAPERS PRACTICE

[10]

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Q6.

(a) $0^2 = 7^2 + 2(-9.8)s$

Use of constant acceleration equation with $v = 0$

M1

$s = \frac{4.9}{19.6} = 2.5$

Correct s

A1

Max Height = $5 + 2.5 = 7.5$ m

ag Adding 5 to get total height

B1

3

(b) $0 = 7 - 9.8t$

Use of constant acceleration equation with $0 = v$

M1

$$t = \frac{7}{9.8} = 0.714 \text{ s}$$

ag *Correct time from correct working*

A1

2

(c) (i) $5.5 = 4.9t^2$

Finding time to fall 5.5 metres

M1

$$t = \sqrt{\frac{5.5}{4.9}} = 1.059$$

Correct time

A1

$$\text{Total Time} = 0.7143 + 1.059 = 1.77 \text{ s}$$

Finding total time

m1

Correct time

A1

4

(c) (ii) $v^2 = 0^2 + 2 \times 9.8 \times 5.5$

*Use of constant acceleration based on
catch at the correct height*

M1

Correct equation

A1

$$v = \sqrt{107.8} = 10.4 \text{ ms}^{-1}$$

Correct speed

A1

3

[12]

Q7.

(a) $v = \int t - \frac{t^2}{5} dt$

Integrating both terms

M1

$$= \frac{t^2}{2} - \frac{t^3}{15} + c$$

Correct integral with or without c

A1

$$v = 0, t = 0 \Rightarrow c = 0$$

Showing c = 0

A1

3

$$v = \frac{t^2}{2} - \frac{t^3}{15}$$

(b) $v(5) = \frac{5^2}{2} - \frac{5^3}{15} = 4.17 \text{ ms}^{-1}$

Substituting t = 5

M1

Correct v

A1

2

(c) $s = \int_0^5 \left(\frac{t^2}{2} - \frac{t^3}{15} \right) dt$

Integrating

M1

$$= \left[\frac{t^3}{6} - \frac{t^4}{60} \right]_0^5$$

Correct expression

A1

*Substitution of two limits or finding c
and substituting t = 5*

m1

$= 10.4 \text{ m}$

Correct distance

sc for only one limit M1A1A1

A1

4

[9]

Q8.

(a) (i) $6^2 = 2^2 + 2 \times a \times 10$

Use of a constant acceleration equation to find a

M1

$$a = \frac{36 - 4}{20} = 1.6 \text{ ms}^{-2}$$

Correct result from correct working

A1

2

(ii) $6 = 2 + 1.6t$

Use of a constant acceleration equation to find t

M1

$$t = \frac{4}{1.6} = 2.5 \text{ s}$$

Correct t from correct working

A1

2

(b) $F - 35 = 65 \times 1.6$

$$F = 104 + 35 = 139 \text{ N}$$

Three term equation of motion

M1

Correct equation

A1

Correct force

A1

3

[7]

Q9.

- (a) $0 = 10 \sin 70^\circ t - 4.9t^2$
Equation for height equal to zero
Solving for t

M1 M1

$$t = 0 \text{ or } t = \frac{10 \sin 70^\circ}{4.9} = 1.918 \text{ s}$$

Correct t
Calculating range

A1 M1

$$R = 10 \cos 70^\circ \times 1.918 = 6.56 \text{ m}$$

Correct range

A1

5

- (b) $-2 = 10 \sin 70^\circ t - 4.9t^2$
Forming equation for vertical motion
when ball lands
LHS correct
RHS correct

M1 A1 A1

$$4.9t^2 - 10 \sin 70^\circ t - 2 = 0$$

$$t = 2.11 \text{ or } -0.193$$

Solving quadratic equation
Correct solution

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$$R = 10 \cos 70^\circ \times 2.11 = 7.22 \text{ m}$$

Calculating range
Correct range

M1A1

7

[12]

Q10.

- (a) (i) $s(10) = 25 - 100 + 150 = 75$
Correct distance

B1

1

$$(ii) \quad v = \frac{t^3}{100} - \frac{3t^2}{10} + 3t$$

Differentiating s
Correct derivative

M1 A1

$$v(10) = 10 - 30 + 30 = 10$$

Correct v

A1

3

$$(iii) \quad a = \frac{3t^2}{100} - \frac{3t}{5} + 3$$

Differentiating v
Correct derivative

M1 A1

$$a(10) = 3 - 6 + 3 = 0$$

Correct a

A1

3

$$(b) \quad h = 10$$

Value of h

B1

$$75 = 100 - k$$

Substituting s = 75 and t = 10

M1

$$k = 25$$

Correct k

A1

3

[10]

Q11.

$$(a) \quad 5 = 32\sin 60^\circ t - 4.9t^2$$

M1

Equation for vertical motion with ± 5

M1

LHS correct

A1

RHS correct

A1

$$4.9t^2 - 32\sin 60^\circ t + 5 = 0$$

$$t = 0.1866 \text{ or } 5.4691$$

Solving quadratic

m1

5.47 seconds

Selecting larger answer from two solutions or obtaining one answer with a reason

A1

5

(b) $32\cos 60^\circ \times 5.469 = 87.5 \text{ m}$

Equation for horizontal motion

M1

Correct range

A1

2

(c) $v_H = 32\cos 60^\circ$

Horiz. component of velocity seen or used

B1

$$v_V = 32\sin 60^\circ - 9.8 \times 5.469 = -25.88$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{16^2 + 25.88^2} = 30.4 \text{ ms}^{-1}$$

Finding magnitude

m1

Correct speed

(Note Max Height = 39.2 m from $t = 2.83$)

A1

5

Q12.

(a) (i) $16^2 = 0^2 + 2 \times a \times 400$

Using constant acceleration equation to find a

M1

$$a = \frac{256}{800} = 0.32 \text{ ms}^{-2}$$

ag *Correct a from correct working*

A1

2

(ii) $16 = 0 + 0.32t$

Using constant acceleration equation to find t

M1

$$t = \frac{16}{0.32} = 50 \text{ seconds}$$

Correct t

A1

2

(b) (i) $30^2 = 16^2 + 2 \times 0.5s$

Using constant acceleration equation to find s

M1

$$s = 30^2 - 16^2 = 644\text{m}$$

Correct s

A1

2

(ii) $644 = \frac{1}{2}(16 + 30)t$

Using constant acceleration equation to find t

M1

$$t = \frac{644}{23} = 28\text{s}$$

Correct t

A1

$$\text{total time} = 28 + 50 = 78 \text{ s}$$

Adding 50 to give total time

A1f.t.

3

[9]

Q13.

(a) $19\mathbf{i} + 13\mathbf{j} = 35\mathbf{i} + 45\mathbf{j} + 8\mathbf{a}$

Constant acceleration equation to find a

M1

$$\mathbf{a} = \frac{19-35}{8}\mathbf{i} + \frac{13-45}{8}\mathbf{j} = 2\mathbf{i} - 4\mathbf{j}$$

Correct equation

A1

ag *Correct a from correct working*

A1

3

(b) $\mathbf{r} = (35\mathbf{i} + 45\mathbf{j})t + \frac{1}{2}(-2\mathbf{i} - 4\mathbf{j})t^2$

Use of constant acceleration equation

M1

Correct i component

A1

Correct j component

A1

3

(c) $\mathbf{r} = (3.5t - t^2)\mathbf{i} + (45t - 2t^2)\mathbf{j}$

Splitting into components correctly

B1

$$35t - t^2 = 300$$

Forming equation for one component

M1

$$t^2 - 35t + 300 = 0$$

Solving quadratic

M1

$$t = 15 \text{ or } t = 20$$

Two correct solutions

A1

$$4t - 2t^2 = 225$$

Forming correct second quadratic

B1

$$2t^2 - 45t + 225 = 0$$

$$t = 7.5 \text{ or } t = 15$$

Solving quadratic for two solutions

M1

$$t = 15 \text{ seconds}$$

Correct final solution

A1

7

[13]

Q14.

(a) (i) $20 \sin 30^\circ t - 4.9t^2 = 0$

Forming equation for time of flight

M1

Correct equation

A1

$$t = 0 \text{ or } t = \frac{20 \sin 30^\circ}{4.9} = 2.04 \text{ (to 3 sf)}$$

Solving quadratic equation

M1

ag *Correct solution from correct working*

A1

4

(ii) $R = 20 \cos 30^\circ \times 2.04 = 35.3 \text{ m}$

Calculation of range

M1

Correct range

A1

(b) $20 \sin 30^\circ t - 4.9t^2 = 2$
Equation to find t at height of 2

M1

Correct equation

A1

$$4.9t^2 - 10t + 2 = 0$$

$$t = 0.2248 \text{ or } t = 1.82$$

Correct times

A1

$$v_y = \sin 30^\circ - 9.8 \times 0.2248 = 7.797$$

Calculating vertical component

M1

Correct vertical component

A1

$$v_x = 20 \cos 30^\circ = 17.32$$

Finding horizontal component

B1

Finding speed from components

M1

$$v = \sqrt{v_x^2 + v_y^2} = 19.0 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Correct speed

A1

8

[14]

Q15.

(a) $s = \frac{1}{2} (40 + 60) \times 4$

Equation based on area of trapezium of rectangle plus two triangles

M1

Correct equation

A1

$$= 200\text{m}$$

Correct final distance

A1

3

(b) $a = \frac{4}{5} = 0.8 \text{ ms}^{-2}$

Attempting to find acceleration

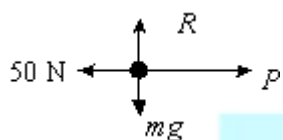
M1

Correct acceleration

A1

2

(c) (i)



Correct force diagram

B1

1

(ii) $P - 50 = 65 \times 0.8$

Three term equation of motion

M1

Correct equation

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A1

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$$P = 102 \text{ N}$$

Correct value

A1

3

[9]

Q16.

(a) $v = 4 - 2e^{-t}$

Differentiating

M1

Correct velocity

A1

2

(b) $a = 2e^{-t}$

Differentiating

M1

Correct acceleration

A1

2

(c) Acceleration decreases, approaching a value of zero

Decreasing

B1

Approaching zero

B1

2

[6]

Q17.

(a) $\mathbf{a} = \frac{1000\mathbf{i} - 5000\mathbf{j}}{2000} = \frac{t}{2}\mathbf{i} - \frac{5}{2}\mathbf{j}$
Using Newton's 2nd law

Correct acceleration

M1

A1

2

(b) $\mathbf{v} = \left(\frac{t^2}{4} + c\right)\mathbf{i} + \left(\frac{-5t}{2} + d\right)\mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$t = 0, \mathbf{v} = 6\mathbf{j} \Rightarrow c = 0, d = 6$

Evaluation of constants

m1

Correct constants and correct final answer

A1

4

$$\mathbf{v} = \left(\frac{t^2}{4} \right) \mathbf{i} + \left(6 - \frac{5t}{2} \right) \mathbf{j}$$

(c) $\mathbf{r} = \left(\frac{t^3}{12} + e \right) \mathbf{i} + \left(6t - \frac{5t^2}{4} + f \right) \mathbf{j}$

Integration

M1

Correct integral with or without constants

A1

$$t = 0, \mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow e = 0, f = 0$$

Evaluation of constants

m1

$$\mathbf{r} = \left(\frac{t^3}{12} \right) \mathbf{i} + \left(6t - \frac{5t^2}{4} \right) \mathbf{j}$$

Correct constants and correct final answer

A1

4

[10]

Q18.

(a) $120 = \frac{1}{2} a \times 50^2$

Use of constant acceleration equation

M1

Correct equation

A1

$$a = \frac{240}{2500} = 0.096 \text{ ms}^{-2}$$

Correct acceleration from correct working

A1

3

(b) $v^2 = 2 \times 0.096 \times 200$

Use of constant acceleration equation

M1

Correct equation

A1

$$v = 6.20 \text{ ms}^{-1}$$

Correct velocity

A1

3

$$(c) \quad 200 = \frac{1}{2} \times 0.096 t^2$$

Use of a constant acceleration equation

M1

Correct equation

A1

$$t = \sqrt{\frac{400}{0.096}} = 64.5 \text{ s}$$

Correct time

A1

3

$$(d) \quad F - 250 \times 9.8 = 250 \times 0.096$$

Three term equation of motion

M1

Correct equation

A1

$$F = 2470 \text{ N (to 3sf)}$$

Correct force

A1

3

[12]

Q19.

$$(a) \quad s = \frac{1}{2} \times 2 \times 4^2 = 16 \text{ m}$$

*Using $s = ut + \frac{1}{2}at^2$ with $u = 0$ or
cao*

M1 A1

2

(b) $v = 2 \times 4 = 8 \text{ ms}^{-1}$

*Using $v = u + at$ with $u = 0$ oe
cao*

M1 A1

2

(c) $100 - 16 = 8t$

Using 100 minus answer to part (a)

M1

$$t = \frac{100 - 16}{8} = 10.5 \text{ s}$$

*Correct equation
10.5 cao*

A1 A1

Total Time = $10.5 + 4 = 14.5\text{s}$

Adding 4 to get 14.5

A1ft

4

[8]

Q20.

(a) $\mathbf{v} = -2\sin\left(\frac{t}{20}\right)\mathbf{i} + 4\cos\left(\frac{t}{20}\right)\mathbf{j}$

*Differentiating
i component correct
j component correct*

M1 A1 A1

3

(b) $\mathbf{v} = 4\mathbf{j}$

Correct velocity at $t = 0$

B1

Travelling north

Travelling north with $\mathbf{v} = n\mathbf{j}$ where $n > 0$

B1

2

(c) $2 \sin\left(\frac{t}{20}\right) = 0$

Equation for t based on 0i

M1

$$\frac{t}{20} = \pi$$

$$t = 20\pi$$

Correct solution

A1

2

[7]

Q21.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$

Summing the forces

M1

$$= 6\mathbf{i} + 5\mathbf{j}$$

Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$

Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$$= 33\mathbf{i} + 23\mathbf{j}$$

Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$

Vector constant acceleration equation to find $\mathbf{r}$

Correct equation

M1 A1

$$= 720\mathbf{i} + 420\mathbf{j}$$

Correct position vector

A1

$$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$$

Finding magnitude

Correct magnitude

M1 A1

5

[10]

Q22.

(a) $20^2 = 12^2 + 2 \times a \times 200$

*Using a constant acceleration equation
with a unknown*

M1

$$a = \frac{400 - 144}{400} = 0.64 \text{ ms}^{-2}$$

Correct equation

Correct result from correct working

A1/A1

3

(b) $20 = 12 + 0.64t$

*Using a constant acceleration equation
with t unknown*

M1

$$t = \frac{20 - 12}{0.64} = 12.5 \text{ seconds}$$

Correct equation

Correct result from correct working

A1/A1

3

(c) (i) $F = 1200 \times 0.64 = 768 \text{ N}$

Correct answer only

B1

1

(ii) $F - 450 = 768$

Three term equation of motion

M1

$$F = 1220 \text{ N}$$

Correct result

Accept 1218

A1

2

[9]

Q23.

(a) $6060 - 600 \times 9.8 = 600a$

Three term equation of motion

M1

$$a = \frac{6060 - 5880}{600} = 0.3 \text{ ms}^{-2}$$

Correct acceleration from correct working

A1

2

(b) $8 = \frac{1}{2} \times 0.3t^2$

*Using a constant acceleration equation
with t unknown*

M1

$$t = \sqrt{\frac{16}{0.3}} = 7.30 \text{ seconds}$$

Correct equation

Correct result

Accept 7.3

A1/A1

3

(c) $v^2 = 2 \times 0.3 \times 8$

*Using a constant acceleration equation
with v unknown and u = 0*

M1

$$v = \sqrt{4.8} = 2.19 \text{ ms}^{-1}$$

Correct velocity

A1

2

[7]

Q24.

(a) $44\mathbf{i} + 28\mathbf{j} = 10(4\mathbf{i} + 2\mathbf{j}) + 50\mathbf{a}$

Equation to find $\mathbf{a}$ using position vector

Correct equation

M1/A1

$\mathbf{a} = 0.08\mathbf{i} + 0.16\mathbf{j}$

Solving for $\mathbf{a}$

Correct $\mathbf{a}$

M1/A1

4

(b) $\mathbf{r} = (4\mathbf{i} + 2\mathbf{j})t + \frac{1}{2}(0.08\mathbf{i} + 0.16\mathbf{j})t^2$

Equation for $\mathbf{r}$

One component correct

M1/A1

$= (4t + 0.04t^2)\mathbf{i} + (2t + 0.08t^2)\mathbf{j}$

Second component correct

A1

3

(c) (i) $96 = 4T + 0.04T^2 \quad 72 = 2T + 0.08T^2$

Forms and solves equation for one component

M1

$T = 20 \text{ or } -120 \quad T = 20 \text{ or } -45$

Correct solutions

A1

Hence $T = 20$

Forms and solves equation for other component

Correct solutions plus correct conclusion

M1/A1

4

(d) $\mathbf{v} = (4\mathbf{i} + 2\mathbf{j}) + (0.08\mathbf{i} + 0.16\mathbf{j}) \times 20$

Finding v at $t = 20$

M1

$$= 5.6\mathbf{i} + 5.2\mathbf{j}$$

Correct v

A1

$$v = \sqrt{5.6^2 + 5.2^2} = 7.64 \text{ ms}^{-1}$$

Finding correct speed

A1

3

[14]



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